Why Statistical Thinking is Important for Data Science

- Recognizes uncertainty and variation in real-world data
- Helps us model and understand underlying processes
- Enables data analysis, explanation of phenomena, and prediction
- Supports evidence-based decision making and solution development

Populations and samples

- A population is every unit we care about
- A sample is the part we actually observe
- Good samples are representative of the population

Variation is the signal

- Every measurement varies
- Statistics separates real differences from noise